

Muhammad Shehroz

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SUMMARY

Aspiring Process Engineer with a background in Metallurgical Engineering and hands-on experience in process optimization and quality control. Contributed to operational efficiency at Amreli Steels by participating in induction furnace operations and continuous casting processes. Observed production and supported continuous improvement initiatives at HinoPak Motors Limited. Eager to leverage engineering expertise to enhance industrial processes and drive innovation.

WORK EXPERIENCE

AMRELI STEELS LIMITED

TRAINEE ENGINEER

Mar 2025 - Present

DHABEJI, PAKISTAN

- Participated in induction furnace floor operations, gaining practical understanding of steel manufacturing processes and improving operational knowledge
- Studied the operation of continuous casting machines, gaining knowledge of process flow, key equipment components, and quality parameters, which enhanced process understanding.
- Contributed to operational efficiency by observing mechanical and electrical maintenance activities.
- Ensured adherence to quality standards by participating in ladle operations and preparation.
- Gained exposure to refractories and their application in steel making, enhancing material application knowledge.
- Gained hands-on experience using SAP H/HANA for material reservation, improving efficiency in inventory management.

PEOPLES STEEL MILLS LIMITED

INTERNEE

Feb 2024 - Feb 2024

Karachi, Pakistan

- Supported Quality Assurance activities by assisting in data analysis using Origin software, contributing to improved product quality.
- Studied and analyzed advanced steelmaking processes, including an introduction to rolling mill operations and quality parameters.
- Assisted in and gained knowledge of destructive and non-destructive testing methods for final products, with exposure to testing rolled steel products.

HINOPAK MOTORS LIMITED

INTERNEE

Mar 2023 - Mar 2023

- Observed and learned about prototype development processes for automobile applications, gaining insights into design and manufacturing techniques.
- Gained experience in using SolidEdge software for drawing automobile parts, enhancing technical drawing skills.
- Observed and analyzed Fiber Reinforced Plastic (FRP) processes, understanding their application in vehicle manufacturing.
- Assisted in vehicle inspections, learning key quality control points and contributing to maintaining high standards.
- Gained insights into Kaizen implementation in the industry, understanding its role in continuous improvement.

EDUCATION

NED University of Engineering and Technology

Bachelors, Metallurgical Engineering

Oct 2020 - Aug 2024

Board of Intermediate Education

Intermediate, Engineering

Sep 2018 - Sep 2020

Board of Secondary Education

Matriculation, Computer Science

Apr 2016 - Aug 2018

CERTIFICATIONS

- ISO 9001-2015- Quality Management System: Alison
- The Seven Basic Quality Control (7-QC) Tools: Alison

PROJECTS

Experimental and Simulation test of projectile impact on ballistic steels for Automobile Applications. Oct 2023 - Aug 2024

- Conducted mechanical testing such as tensile and hardness test on Ramor 500/550 to obtain simulation input parameters.
- Simulated projectile impact on Ramor 500/550 steels for automotive safety, using mechanical testing data.
- Developed explicit dynamic finite element models in Ansys AutoDyn to simulate impact on Ramor 500/550.
- Implemented Johnson Cook material model calibrated with Ramor 500/550 test data.
- Analyzed and compared simulation results with the mechanical testing behavior of Ramor 500/550.

Development of thin films for packaging applications.

Mar 2023 - Apr 2023

- Developed thin films using Poly vinyl Alcohol (PVA) and Toluene.
- Performed Thermo Gravimetric Analysis (TGA) and Differential Scanning Calorimetry (DSC) to study their properties.

VOLUNTEERING & LEADERSHIP

NED University of Engineering and Technology
Community Service Coordinator

May 2022 - Jun 2022

SKILLS

- Communication, Excel for data analysis, Gemba, Kaizen, Microsoft Office, Microsoft Office package, Origin software for data analysis, SAP, Time Management, Manufacturing , Quality Assurance